



HASHIKO

{File Integrity Monitor}

Jackie Ma | A00889988

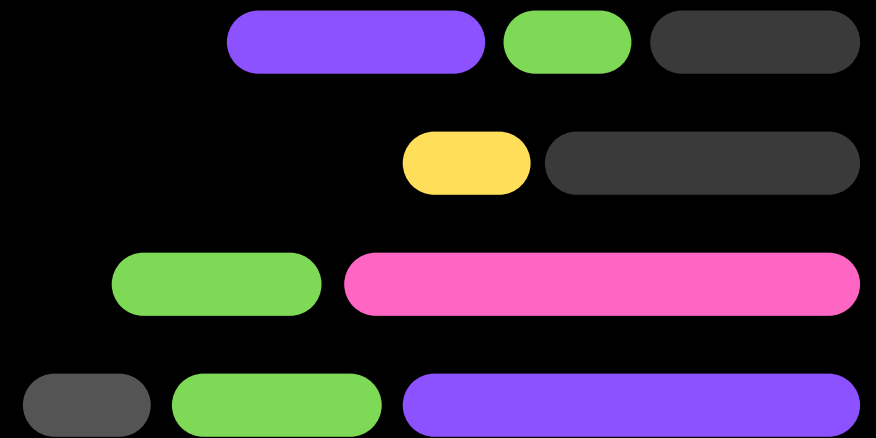
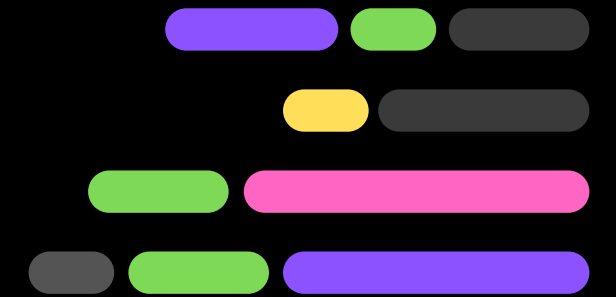




TABLE OF CONTENTS



01

SYSTEM OVERVIEW

What is Hashiko?

02

MILESTONES

Milestones 3, 4 and 5

03

DEMO

Live Demo

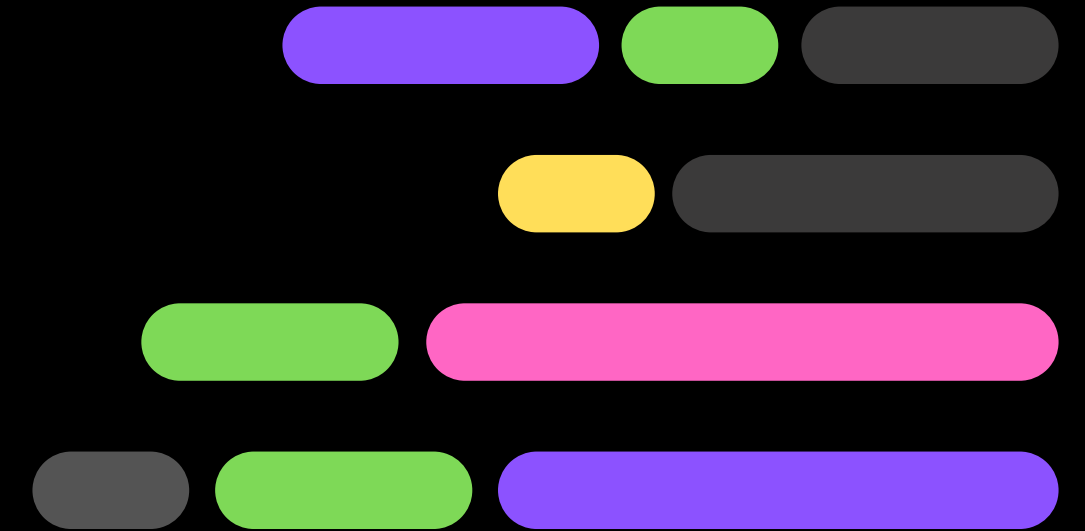


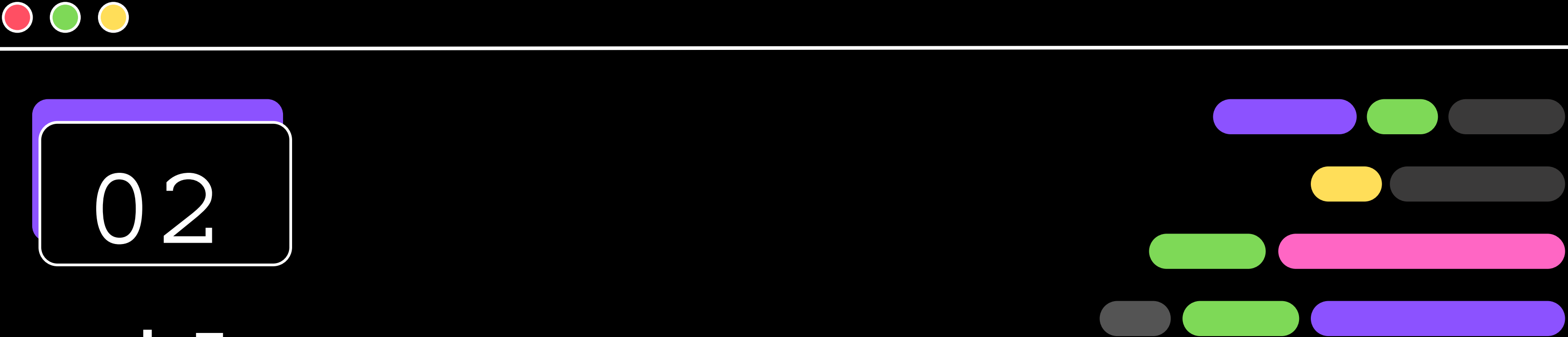
01

System Overview

What is Hashiko?

- File integrity monitoring tool for Linux systems
- Detects unauthorized changes to critical system files
- Agent computes SHA-256 hashes and sends reports to a central backend
- Web UI displays hash history and alerts
- Offline queue retries failed reports when the server is unreachable
- Deployed on Render with PostgreSQL on Supabase



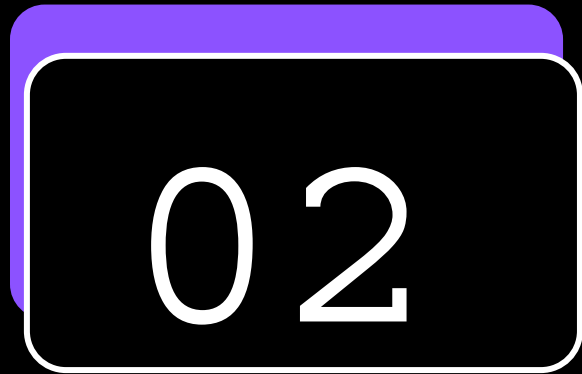


02

Milestone 1-2 Recap

Agent Hashing and Offline Queue

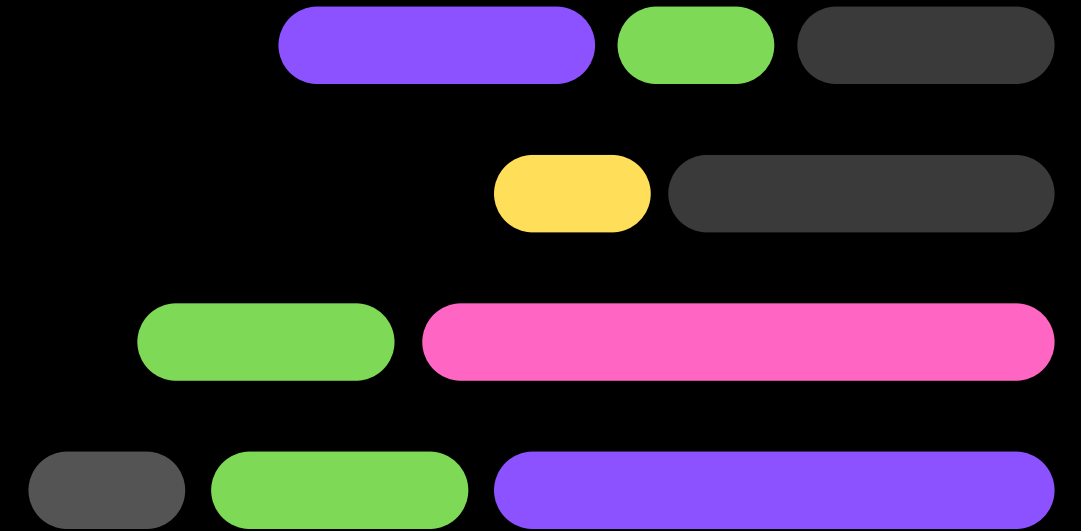
- Hashes of five directories: /boot, /usr/bin, /usr/sbin, /etc and /root
- Backend rewritten in Flask with a PostgreSQL database
- Hash records stored and displayed in local web UI
- Saves failed reports to locally and retries when the server is reachable

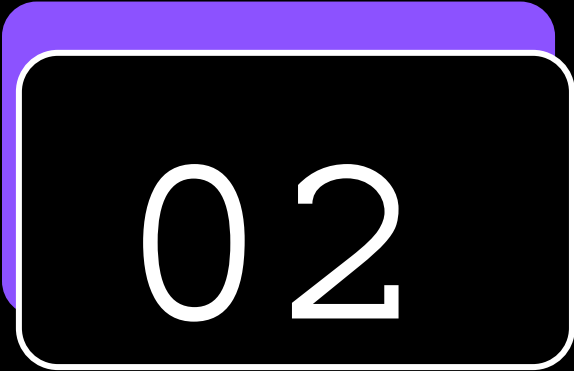


Milestone 3

Hash Comparison, Alerts and Login

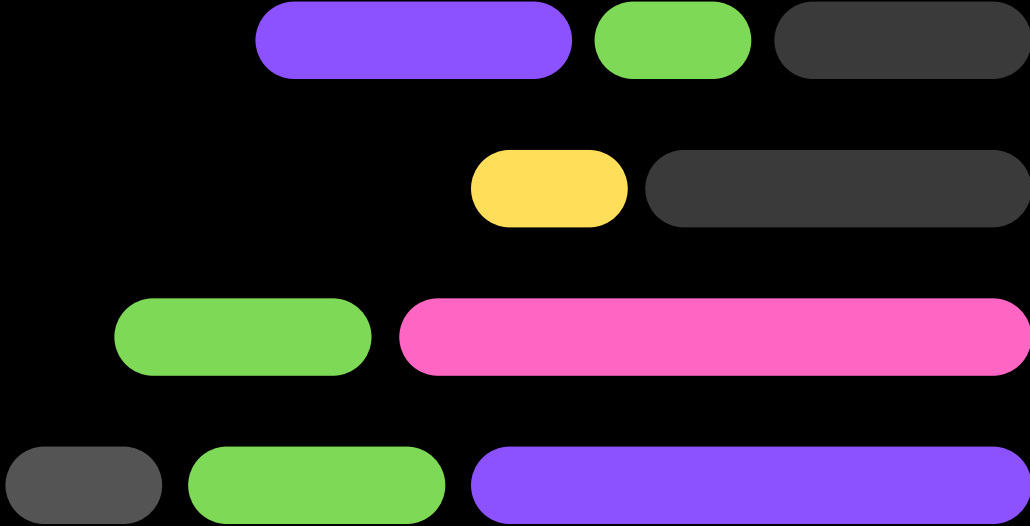
- Hash comparison: each record checked against the previous one per agent
- Changes flagged and surfaced in an alert panel
- Alerts limited to the last 24 hours
- Login added to restrict all data access





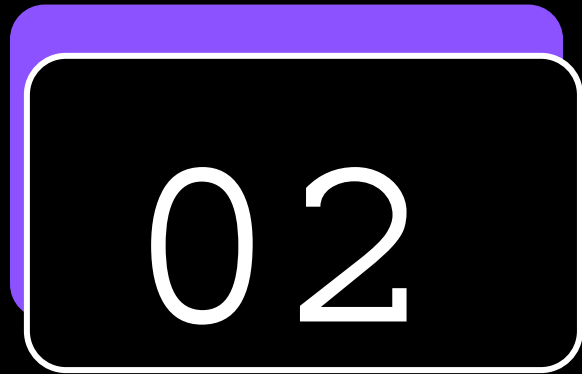
02

Milestone 4

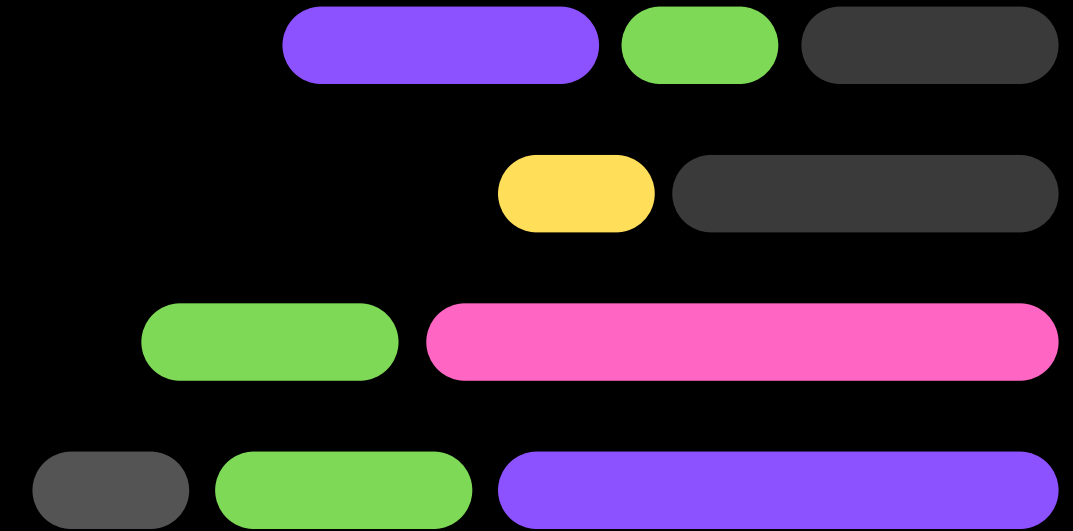


Security and Deployment

- Unique agent ID: `user@hostname (machine-id)`
- One-time registration token exchanges for a persistent API key
- Per-agent watch config managed from the web UI
- HTTPS enforced on agent, SSL on database, HSTS header, secure cookies
- Render provides TLS automatically



Milestone 5

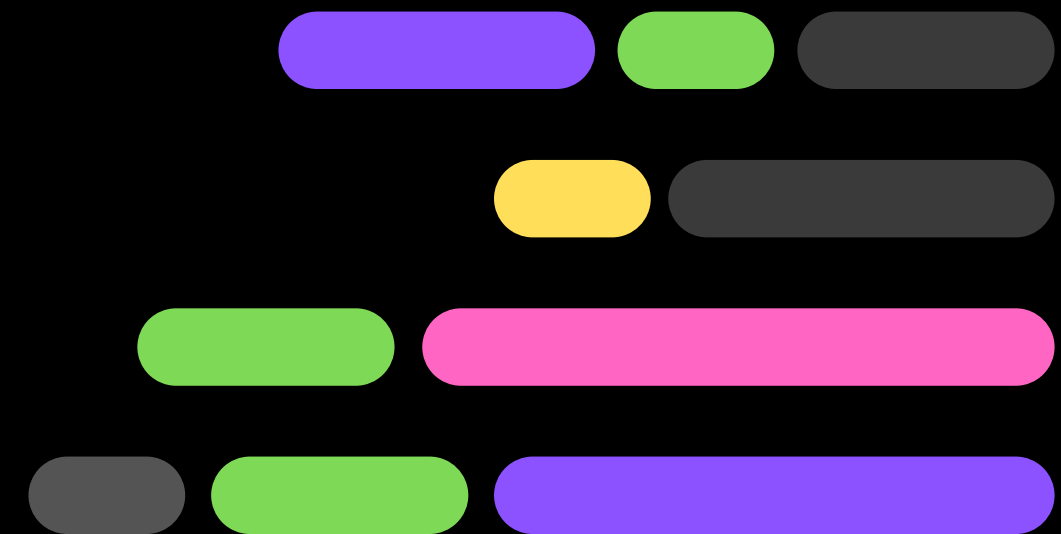


Testing and CI

- 9 Go tests: skip logic, hash correctness, large file filter, change detection
- 23 Python tests: change detection algorithm and all API endpoints
- Tests run against a real PostgreSQL database
- GitHub Actions runs both jobs in parallel on every push
- Dismiss alerts
- Set new baseline
- Hash individual files within directories



DEMO



CONCLUSION



Milestone	Key Deliverables
3	Hash comparison, alerts, login
4	Registration, per-agent config, security, deployment
5	Per-file hashing, size threshold, testing, CI

- System is live and monitoring in production
- Each milestone built directly on the previous one

THANK YOU

Any questions?

